GRADE 11 TECHPRO ACTIVITY SHEET
CSS3 GRADIENTS

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SECTION: 11 - Bell

Learning Objectives

- Explain how CSS gradients create smooth transitions between colors.
- Create linear gradients with different directions and color stops.
- Create radial gradients that spread outward from a central point.
- Create conic gradients that rotate around a center point.
- Use gradients as backgrounds for webpage sections and cards.
- Build and test a simple TechPro webpage in Visual Studio Code.

Instructions

- Create a folder named CSS_Gradients_Activity.
- Open Visual Studio Code, then choose File > Open Folder and select your folder.
- Create a file named index.html.
- Type each code example into index.html and save it.
- Open index.html in a browser. If you have the Live Server extension, right-click the file and choose Open with Live Server.
- Change the colors, directions, and sizes as instructed. Observe the result after each change.

Quick Guide: Three Types of CSS Gradients

A gradient is an image generated by CSS. It can be used anywhere a background image is accepted.

• linear-gradient(): colors transition along a straight line.

• radial-gradient(): colors transition outward from a central point in an ellipse or circle.

• conic-gradient(): colors transition around a center point, like slices on a color wheel.

PART A — Create the Starter Page

Enter this starter code in index.html. You will add more CSS and HTML in the next parts.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Gradients Activity</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 24px;
```

```
      background: #f5f5f5;
      color: #222;
    }
    h1 { color: maroon; }
    .demo {
      min-height: 120px;
      padding: 20px;
      margin: 14px 0;
      border-radius: 12px;
      color: #222;
    }
  </style>
</head>
<body>
  <h1>My CSS Gradients Activity</h1>
  <p>Gradients blend colors to make webpages more engaging.</p>
  <div class="demo">My first gradient will go here.</div>
</body>
</html>
```

PART B — Linear Gradients

A linear gradient blends colors along a line. Add this rule inside the <style> section:

```
.demo {
  min-height: 120px;
  padding: 20px;
  margin: 14px 0;
  border-radius: 12px;
  background: linear-gradient(to right, maroon, pink);
}
```

Try these variations one at a time. Replace the background value and refresh the browser.

```
background: linear-gradient(to bottom, navy, skyblue);
background: linear-gradient(45deg, orange, gold);
background: linear-gradient(to right, maroon, white, gold);
```

Activity:

- Change the direction to to left. What changes?
- Use three colors of your choice in one gradient.
- Try a color stop, for example: linear-gradient(to right, maroon 20%, pink 80%).

PART C — Radial Gradients

A radial gradient spreads from a center point. Add a new class so you can compare it with the linear gradient.

```
.radial-demo {
  min-height: 150px;
  padding: 20px;
  margin: 14px 0;
  border-radius: 12px;
  background: radial-gradient(circle, white, maroon);
}
```

Add this HTML below your first :

```
<div class="radial-demo demo">Radial Gradient</div>
```

Try these values:

```
background: radial-gradient(circle, yellow, orange, red);
background: radial-gradient(ellipse at top left, white, teal);
background: radial-gradient(circle at center, gold 10%, maroon 70%);
```

Activity:

- Change circle to ellipse. Compare the shape of the gradient.
- Change the center position to top left or bottom right.
- Choose colors that have enough contrast for any text placed over the gradient.

PART D — Conic Gradients

A conic gradient rotates color transitions around a center point. It can create a color-wheel effect.

```
.conic-demo {
  width: 180px;
  height: 180px;
  margin: 18px auto;
  border-radius: 50%;
  background: conic-gradient(red, yellow, lime, aqua, blue, magenta, red);
}
```

Add this HTML below your other demo elements:

```
<div class="conic-demo" aria-label="Conic gradient color wheel"></div>
```

Try these variations:

```
background: conic-gradient(maroon, pink, gold, maroon);
background: conic-gradient(from 90deg, red, yellow, lime, blue, red);
background: conic-gradient(red 0deg 90deg, yellow 90deg 180deg,
                           lime 180deg 270deg, blue 270deg 360deg);
```

Activity:

- Change the starting angle using from 90deg.
- Create a four-color wheel using red, yellow, green, and blue.
- Adjust the width and height. Keep them equal to maintain a circle.

PART E — Compare the Three Gradient Types

Create three labeled sections on your page: Linear, Radial, and Conic. Each should have a different gradient.

```
<section class="card linear-card">
  <h2>Linear Gradient</h2>
  <p>Color changes along a straight line.</p>
</section>

<section class="card radial-card">
  <h2>Radial Gradient</h2>
  <p>Color spreads outward from a center point.</p>
</section>

<section class="card conic-card">
  <h2>Conic Gradient</h2>
  <p>Color changes around a center point.</p>
</section>
```

```
.card {
  padding: 22px;
  margin: 16px 0;
```

```
  border-radius: 14px;
  min-height: 90px;
}
.linear-card { background: linear-gradient(120deg, #800020, #f4c2c2); }
.radial-card { background: radial-gradient(circle, #fff4cc, #d97706); }
.conic-card { background: conic-gradient(from 45deg, #800020, #f4c2c2, #f5c542, #800020); }
```

Note: If you use the same class name more than once, the same CSS rule will apply to every element with that class.

PART F — Mini Project: TechPro Gradient Showcase

Create a one-page webpage titled “TechPro Gradient Showcase.” Your page must include:

- A main heading and a short introduction.
- One linear-gradient section with at least two colors.
- One radial-gradient section with at least two colors.
- One conic-gradient circle or color wheel.
- Rounded corners, spacing, and readable text.
- A maroon color somewhere in your design.
- A short label explaining each gradient type.

Suggested design CSS:

```
.page-title {
  text-align: center;
  color: maroon;
}
.showcase {
  padding: 24px;
  border-radius: 14px;
  margin: 18px 0;
  color: #222;
}
.linear { background: linear-gradient(to right, #800020, #f7d6df); }
.radial { background: radial-gradient(circle, #fff, #d4a017); }
.conic {
  width: 160px;
  height: 160px;
  border-radius: 50%;
  margin: 20px auto;
  background: conic-gradient(#800020, #f7d6df, #d4a017, #800020);
}
```

PART G — Challenge Task

Design a “Future in Technology” banner using a linear gradient. Add a radial-gradient highlight and a conic-gradient badge or circle. Choose your own colors, but keep the page readable and professional.

Challenge requirements:

- Use at least three colors in one gradient.
- Use a direction or angle in a linear gradient.
- Use a shape or position setting in a radial gradient.
- Use from with a degree value in a conic gradient.
- Add a brief description of where gradients could be useful in a real website.

Check Your Work

- ☐ Created CSS_Gradients_Activity and index.html.
- ☐ Opened the page in a browser and checked that it displays.
- ☐ Tested linear-gradient() with different directions.
- ☐ Tested radial-gradient() with different shapes or positions.
- ☐ Tested conic-gradient() with a starting angle or color stops.
- ☐ Completed the TechPro Gradient Showcase.
- ☐ Saved and tested the final work.

Review Questions

1. What is a CSS gradient?

Answer: a gradient displays a smooth transition between two or more colors

2. What is the main difference between linear-gradient() and radial-gradient()?

Answer: linear gradient splits the colors along a straight line, radial spreads the color from a center point

3. What does conic-gradient() do?

Answer: gradient with color transitions rotated around a center point

4. In linear-gradient(to right, maroon, pink), what does to right control?

Answer: the phrase to right controls the direction of the gradient line meaning the first color value will be displayed on the right first

5. What is a color stop in a gradient?

Answer: color stop is the specific point along a gradient line where a color reaches its full pure intensity

6. What does circle do in radial-gradient(circle, white, maroon)?

Answer: the gradient starts with white and smoothly transitions outward to maroon

7. What does from 90deg change in a conic gradient?

Answer: rotates the starting angle of the gradient 90 degrees clockwise from its default position

8. Name one way gradients can improve a webpage design.

Answer: in can help webpage design by adding depth with the colors or even guiding the user to certain parts of the page

Quick Review

linear-gradient() = blends colors along a straight line.

radial-gradient() = blends colors outward from a center point.

conic-gradient() = blends colors around a center point.

Color stops = specify where a color begins or ends in the gradient.

Gradients can be used as CSS backgrounds for cards, banners, buttons, and decorative shapes.

References

W3Schools — CSS Gradients: https://www.w3schools.com/css/css3_gradients.asp

W3Schools — CSS Radial Gradients: https://www.w3schools.com/css/css3_gradients_radial.asp

W3Schools — CSS Conic Gradients: https://www.w3schools.com/css/css3_gradients_conic.asp